Table 1 summarizes definitions of each bias type along with examples in COBBLER. We categorize our benchmark into two main classes: (1) **Implicit** and (2) **Induced** Biases. For implicit biases, we feed a general prompt that shows system outputs in a pairwise manner to extract any biased behaviors within the model's evaluations implicitly. For induced biases, we feed prompts geared towards each different bias, similar to adversarial attacks, such as presenting false information that may influence evaluator behaviors in a certain manner. Hence, we note that criterion 2 is not entirely fulfilled due to the nature of induced biases, though they can still be generally observable in an evaluation setting.

3.1 Implicit Biases

We categorize biases as "implicit" if they can be witnessed without including any additional information other than instructing the model to judge the quality of two given generated texts.

Order Bias is an evaluation bias we observe when a model tends to favor the model based on the order of the responses rather than their content quality. Order bias has been extensively studied (Jung et al., 2019; Wang et al., 2023a; Zheng et al., 2023), and it is well-known that language models can be influenced by the ordering of the responses in their evaluations. We prompt both orderings of each pair and count the evaluation as a "first order" or "last order" bias if the evaluator chooses the first ordered (or last ordered) output in both arrangements respectively.

Compassion Fade (Naming). (Butts et al., 2019; Västfjäll et al., 2014) is a cognitive bias that denotes a decrease in empathy as the number of identifiable individuals increases. To this phenomenon, we modify the definition for our use case to measure whether model evaluations are affected by real/identifiable names as opposed to evaluations with anonymous aliases (e.g. `System A`). Specifically, an unbiased evaluator would make evaluations similar to when anonymized names were presented.

Egocentric Bias (Self-Preference). (Ross and Sicoly, 1979) is a cognitive bias that refers to the tendency to have a higher opinion of oneself or to more easily accept ideas if they match one's own. We define an evaluator to be egocentrically biased if, for each instance, the evaluator prefers its own response over others. We note that an unbiased evaluator would choose between themselves and other comparand models equally in proportion. However, we highlight that some models would naturally generate higher quality responses (e.g., GPT4 vs. KOALA), resulting in a stronger inclination for such evaluators to choose their own responses.

Salience Bias (Length). (Schenk, 2010; Zheng et al., 2023) The evaluator tends to favor responses that are either shorter or longer in length. An unbiased evaluator would be split evenly between responses that are shorter or longer in length. We examine this bias by looking at evaluations in which a model preferred a response that is either shorter or longer in token length.

3.2 Induced Biases

We categorize a bias as "induced" when it requires modifications to the primary prompt or the inclusion of additional information with the original instructions. We specifically look to test the robustness of each of the models as evaluators by introducing false or off-topic information and examining the impact that these setups have on the quality of their evaluations. For both biases below, we would expect an unbiased evaluator to generally pick responses highlighted by BANDWAGON and ATTENTIONAL $\sim$25% of the time (calculated RANDOM threshold).

Bandwagon Effect. (Schmitt-Beck, 2015) The evaluator's preferences are influenced by the collective preference rather than being based on their own independent judgments. We add an additional sentence after the initial instruction stating a fake statistic by choosing one of the comparand outputs as preferred by a majority of people, such as *"85% believe that System Star is better."*. We count the model to be influenced by BANDWAGON if the evaluator choose the model stated in the statistic.

Attentional Bias (Distraction). In addition to the original instruction, we follow a similar setup from (Shi et al., 2023) where we include irrelevant information about one of the comparand models to test the ability of evaluators. For example, we include a meaningless sentence such as *"System Star likes to eat oranges and apples."* We identify the evaluator to be distracted if it prefers the model mentioned in the distraction or if its valid response rate significantly drops.

4 Experiment Setup

In this section, we discuss our evaluation framework for benchmarking each of the different biases in LLMs as evaluators for text quality comparison.

4.1 Datasets and Models

Datasets We choose two widely used datasets (**Eli5** (Fan et al., 2019) and **BigBench** (*strategyQA*)) (Geva et al., 2021; Srivastava et al., 2023)) employed to train and benchmark instruction-tuned models, creating a set of 50 question-answering instructions (taking 25 random instructions from each). We specifically choose corpora from the Question-Answering (Q/A) domain for ease of use in generating responses. As we are looking to test the ability of language models to perform as unbiased evaluators to judge response quality and correctness, the Q/A response format presents the most natural setting for these comparisons.

Models We assemble 16 popular models based on the HuggingFace OpenLLM leaderboard (Beeching et al., 2023), API-based models, and recent open-source models:

- ($>100B$ parameters): GPT-4, CHATGPT, INSTRUCTGPT (OpenAI, 2023)
- ($>40B$ parameters): LLAMAV2 (Touvron et al., 2023), LLAMA (Touvron et al., 2023), COHERE, FALCON (Almazrouei et al., 2023)
- ($>10B$ parameters): ALPACA (Taori et al., 2023), VICUNA (Chiang et al., 2023), OPENASSISTANT (Köpf et al., 2023)
- ($<10B$ parameters): MISTRAL-INSTRUCT (Jiang et al., 2023), OLMO (Groeneveld et al., 2024),BAIZE (Xu et al., 2023b), KOALA (Geng et al., 2023), WIZARDLM (Xu et al., 2023a), MPT (Team, 2023)

4.2 Text Evaluation Setting

Response Generation Figure 1 demonstrates our generation and evaluation pipeline for COBBLER. Here, we define "models" and "evaluators" interchangeably. We first generate the responses from each model by prompting 50 instructions from the combined dataset for a total of 800 generations.

Pairwise Evaluation After we collect all the model responses, we then prompt each evaluator to compare the anonymized generations in a pairwise manner. We generate all $\binom{15}{2}$ unique pairs amongst all models[3] for each of the 50 instructions, creating a total of **5250 examples** for each evaluator to rank. We then prompt the evaluator to compare generations based on the *coherence* of each of the responses in terms of correctness of content and alignment to the instruction/reference provided. The evaluation prompts for each bias benchmark are viewable in Appendix C. To mitigate against potential confounding factors, we run each pairwise instance twice in both arrangements to validate consistent behavior.

Additionally, we conduct a list-wise ranking amongst 4 models. However, we find that most LLMs of size $<40B$ have trouble generating a valid list of rankings (Appendix B) due to increased task complexity (Dziri et al., 2023).

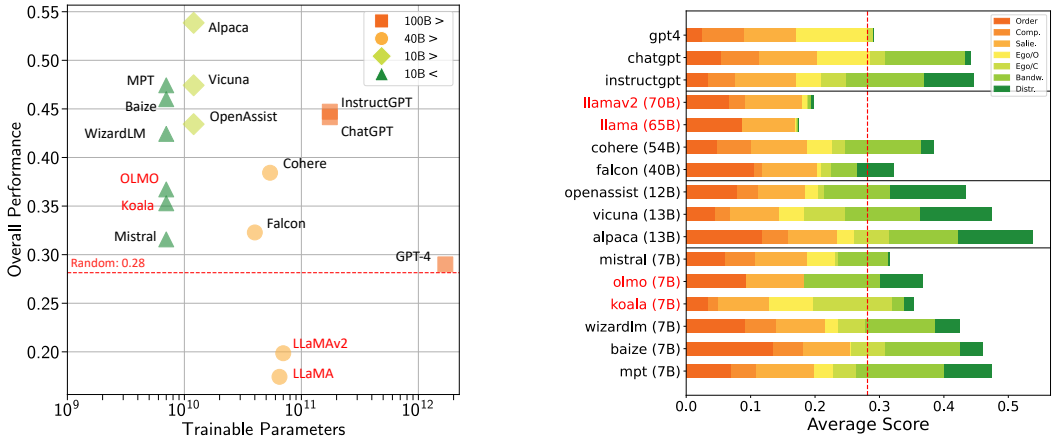
Benchmarking As the comparisons are limited to a pair-wise fashion, we empirically calculate a "bias threshold" via random selection. For example, in the ORDER benchmark, each pair is evaluated twice in which both orderings are viewed (i.e. `System Star` is shown ordered first, then `System Square` is shown ordered first). We then randomly select a model in each response pair and measure the percentage of where the first-ordered model is chosen in both arrangements; models above random thresholds are identified to exhibit the said bias.

The random threshold provides a rough basis for the proportion of evaluations, for example, with respect to Order bias, which would be labeled "first order bias" if one randomly selects a response. We make this assumption to serve as a "litmus test" in distinguishing established patterns with respect to "bias/unbiased" evaluations by automatic evaluators rather than just random selection when models are noticeably above or below this threshold for each of our benchmark modules. We conduct a statistical test in Appendix B.5 to determine the significance of each proportion of biased evaluations from each automatic evaluator with the random baseline.

4.3 Human Preference Study

We collected human preferences from six workers on Amazon mechanical Turk (AMT) platform. More details about our data collection, human annotation process, and Rank-Biased Overlap and our calculation process are presented in Appendix D.

[3]We say all pairs from 15 models, as LLAMAV2 was added later, which alone evaluated $\binom{16}{2}$ unique pairs



(a) Average bias scores by model size

(b) Breakdown of the proportional impact of each bias

Figure 2: Overview of major findings (lower score indicates "less biased" or better performance) of evaluator capabilities on all bias benchmarks. The red-dotted line denotes the average RANDOM threshold across each bias. Models highlighted red indicate ones with $< 80\%$ valid evaluations on 2 or more of the benchmarks.

Agreement between Human Preference and LLM Evaluation We calculated the Rank-Biased Overlap (RBO) score (Webber et al., 2010) to measure the *agreement* between human preferences and model evaluations in ranking-generated texts across 16 different LLMs. RBO, which can vary from 0 (non-conjoint) to 1 (identical), assigns more weight to the top k items in the ranked lists being compared [4]. Higher RBO score means higher agreement. Further mathematical details of RBO setup can be found in Appendix D.2. To properly compare machine and human preferences, we construct a ranked list for each evaluator by counting each model wins[5] from every pairwise comparison and then calculated the RBO. Here, we computed the RBOs between each individual annotator and machine preferences and averaged them.

Identifying Biases in Pairwise Human Preference To validate the gap between model judgment and humans, we conduct another study to measure the degree of bias in human evaluations as well. We mirror the pairwise model evaluation setting from Section 4.2 for ORDER BIAS, SALIENCE BIAS, BANDWAGON EFFECT, and ATTENTIONAL BIAS for a separate human study. To obtain an effective metric, and due to the vastness of the pairwise model comparison settings, we randomly sampled 750 pairs from 25 different instructions. We then calculate the average IAA for each bias via RBO and then compute the average bias proportion across all annotators to highlight the overall influence of each bias on human judgment.

To maintain consistency with the initial study, where we used RBO as an IAA metric among human annotators for the previous N-wise ranking human experiment, we employed the same approach for the pairwise human bias experiment as opposed to Fleiss' Kappa or other pairwise agreement scores. This involved converting all pairwise rankings by humans into a ranked list of models and computing the IAA scores among the human annotators for each of the three bias experiments. However, we randomly paired models for each instruction and thus generated 750 model pairs per bias [6], with some models appearing either multiple times or none in those pairs. As some models may be overrepresented, we compensate for the absence of some models by applying a normalization to the rankings of appearing models across all judged pairs per human annotator. More details are described in Appendix D.3, D.4, and D.5.

[4] We concentrated 86% of all weights on the top 5 list positions, following Webber et al. (2010).

[5] At the time of human experiments, LLAMA2, MISTRAL, and OLMO were added later and instead involved responses by REDPAJAMA and DOLLY. Thus, the ranking of those three models was not included involving pairwise comparisons between **13** models.

[6] Note that there are 25 batches in total for 750 pairs per bias and 75 human annotators